



Q. No	Question	CO-K Level	Max. Mark
9 (a)	Explain the functions of an OS.		12
(OR)			
(b)	Explain the different structures of an Operating System with suitable diagrams. Discuss the advantages and disadvantages of each structure.		12
10 (a)	Describe the types of system calls and the methods used to pass their parameters.		12
(OR)			
(b)	Explain the concept of threads and discuss the different threading models. Differentiate between a process and a thread, with suitable examples and a brief case study.		12

11 (a)	Explain the FCFS, preemptive and non-preemptive versions of Shortest Job First (SJF), and Round Robin (time slice = 2) scheduling algorithms with Gantt charts for the four processes given below. Compare their average turnaround time and average waiting time. <table><tr><th>Proce ss</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>0</td><td>10</td></tr><tr><td>P2</td><td>1</td><td>6</td></tr><tr><td>P3</td><td>2</td><td>12</td></tr><tr><td>P4</td><td>3</td><td>15</td></tr></table>	Proce ss	Arrival Time	Burst Time	P1	0	10	P2	1	6	P3	2	12	P4	3	15		12
Proce ss	Arrival Time	Burst Time																
P1	0	10																
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Competency Level Analysis:

Competence level	Revised Blooms' Taxonomy	Question No.	Marks	Contribution in %
K1	Remember	1, 4, 9a, 10b	26	30%
K2	Understand	2, 3, 6, 11a	17	20%
K3	Apply	8, 10a	15	17%
K4	Analyze	5, 7, 11b	16	19%
K5	Evaluate	9b	12	14%
K6	Create	-	-	-
Total			86	100

Course Outcome	Marks	Contribution in %
CO3	29	34%
CO4	29	34%
CO5	28	32%
Total	86	100

Prepared By Course Coordinator Verified By IQAC Approved By HoD